

William Wei

(437)-445-3268 | wjw9857@nyu.edu | [linkedin.com/in/williamwei0/](https://www.linkedin.com/in/williamwei0/) | github.com/william-wei0 | Ajax, Ontario

EDUCATION

New York University

May 2025

Bachelor of Science in Biomolecular Science

GPA: 3.95/4.00

- **Minors:** Computer Science and Mathematics.

Relevant Courses: Object Oriented Programming, Data Structures & Algorithms, Digital Logic and State Machine Design, Linear Algebra & Differential Equations, Calculus I-III, Statistics

TECHNICAL SKILLS

- **Languages:** JavaScript, TypeScript, Python, C++, SQL, HTML, CSS.
- **Web Technologies/Frameworks:** React, Node.js, Express.js, PostgreSQL, Prisma ORM
- **Data & ML:** PyTorch, NumPy, Pandas, SciPy, scikit-learn, Matplotlib, OpenCV
- **Developer Tools:** Git, Github Actions, Vitest, Docker, AWS (Amplify, Elastic Beanstalk, RDS)

EXPERIENCE

Machine Learning Research Assistant

Sept 2024 – March 2026

New York University, Smart Microphysiological Systems for Precision Medicine

New York, NY

- Designed an LSTM model in Pytorch to forecast 3-year pancreatic cancer responses, enabling personalized drug recommendations from predicted health recovery scores/survival graphs, achieving an AUC 0.89 on validation.
- Developed an image pipeline to automatically extract 15 features from 80k+ time-lapse images, producing a NumPy/Panda dataset with 900k+ data points.
- Automated statistical tests and data visualizations with Scipy/Matplotlib, reducing manual analysis time by 80%.
- Containerized models with Docker to ensure consistent deployments and results across environments.

Physical Chemistry I Course Assistant

Jan 2025 – May 2025

New York University

New York, NY

- Developed a full stack website (physicalchem.org) using React/Express.js/PostgreSQL/TypeScript with interactive 3D simulations of quantum mechanics to help 100+ students learn physical chemistry.
- Built a RESTful API with Node.js and Express.js, implementing role-based access control, JWT authentication, rate limiting and CORS policies to improve API security and protect against unauthorized access.
- Implemented a PostgreSQL database with Prisma ORM to track student progress for 100+ student accounts.
- Designed a responsive and interactive frontend UI for simulations using React/Tailwind/Vite.
- Automated frontend/API testing with Vitest/Postman to validate authentication & ensure reliable deployments.
- Deployed using AWS cloud hosting with Amplify/Elastic Beanstalk/Amazon RDS to ensure 24/7 availability.

PROJECTS

LLM Research Paper Summarization and Translation Tool | JavaScript, React, LLMs

- Built Electron desktop app with IPC architecture to isolate API calls, so API keys are only stored in OS keychain.
- Created light-weight RAG pipeline with OpenAI embeddings/cosine similarity to add domain-specific context.
- Implemented PDF text extraction to process PDFs & added hallucination checks to ensure accurate summaries.

YOLO-based Crystal Detection Algorithm | Python, YOLO, OpenCV

- Fine-tuned a YOLO object tracking model with CUDA to automate measurements of crystal size/shape from images, reducing data acquisition times by 32% and an accuracy within 15% of expected values.
- Identified & discussed requirements from researchers and managed version control/documentation on Github.

2D Tile-Based Game Engine | C++, OpenGL

- Architected an entity system, enabling shared and overridable behavior across player, enemy, projectile classes.
- Encapsulated GLSL shader compilation and managed low-level rendering with UV mapping/matrix transforms.